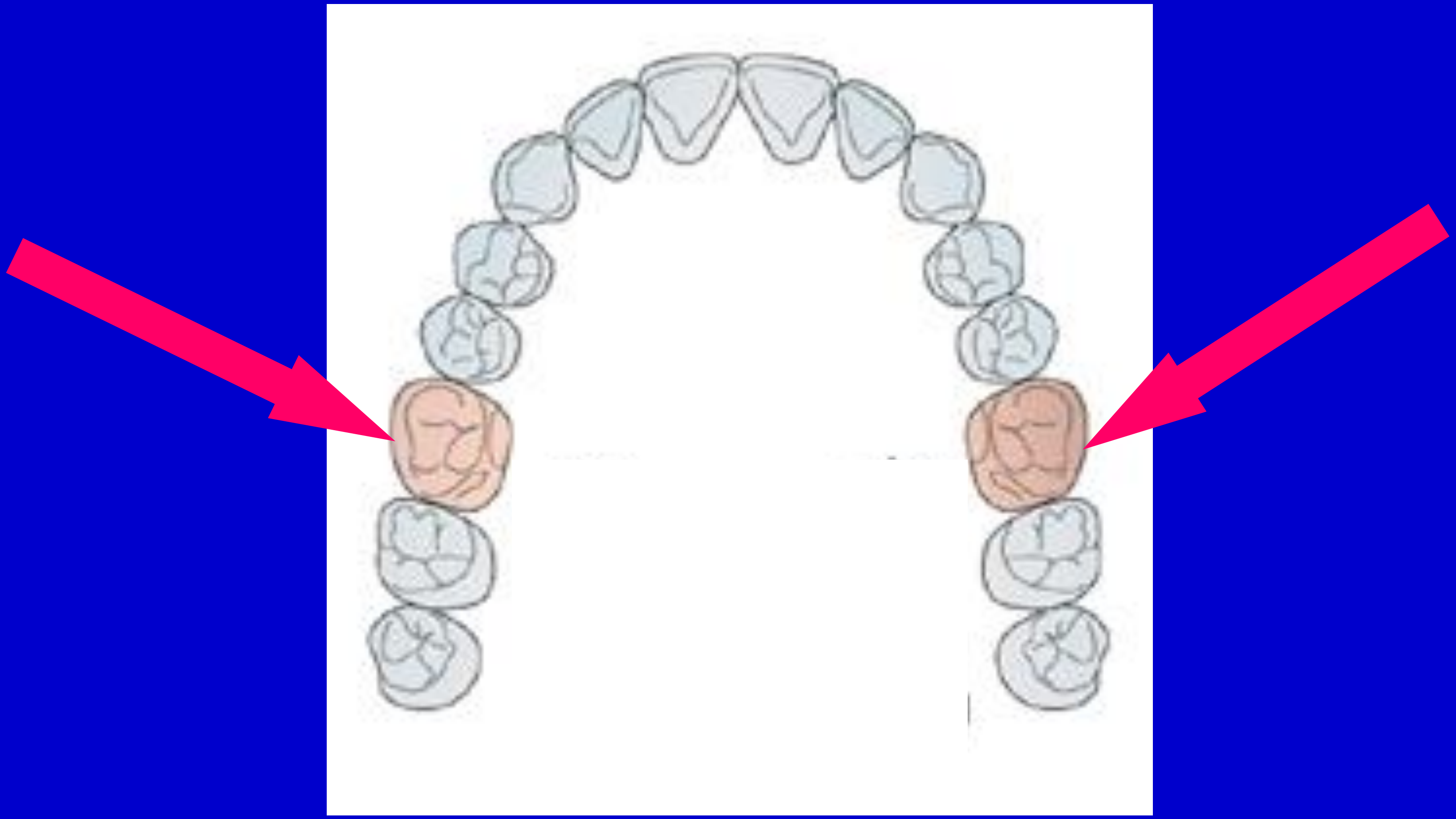
بسم الله الرحمن الرحيم

Permanent Molars

- They are 12 in number
- Function: crushing and grinding of the food
- Multicusped teeth with a wide occlusal surface
- Upper molars have 3roots, lower molars have 2
- Occlusal outline: upper; rhomboidal or heart
  lower; hexagonal or rectangular
- Lower molars' crowns inclined lingually
- No predecessors
- All molars occlude with 2opposing teeth except upper8

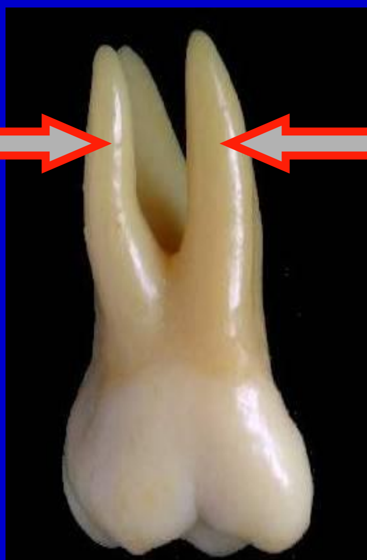
SURFACE ANATOMY OF PERMENANT MAXILLARY MOLARS

Maxillary permanent 1ST Molar

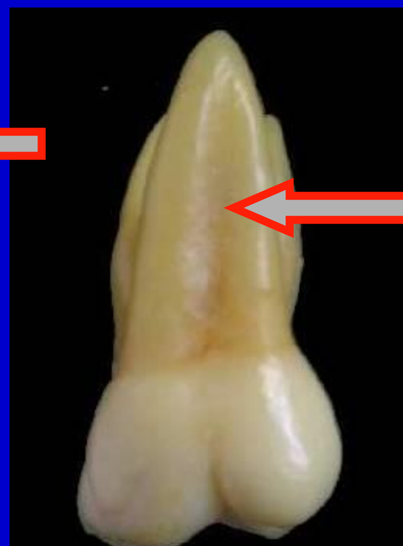


No. of surfaces:

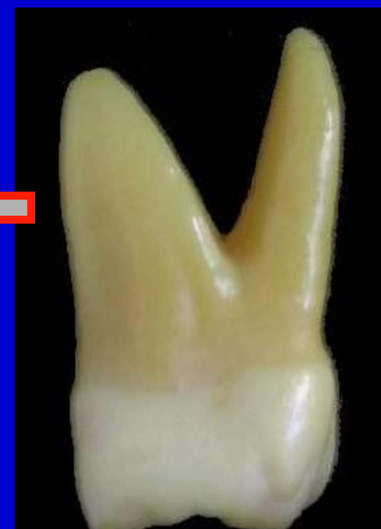
It has five surfaces



BUCCAL



PALATAL



MESIAL

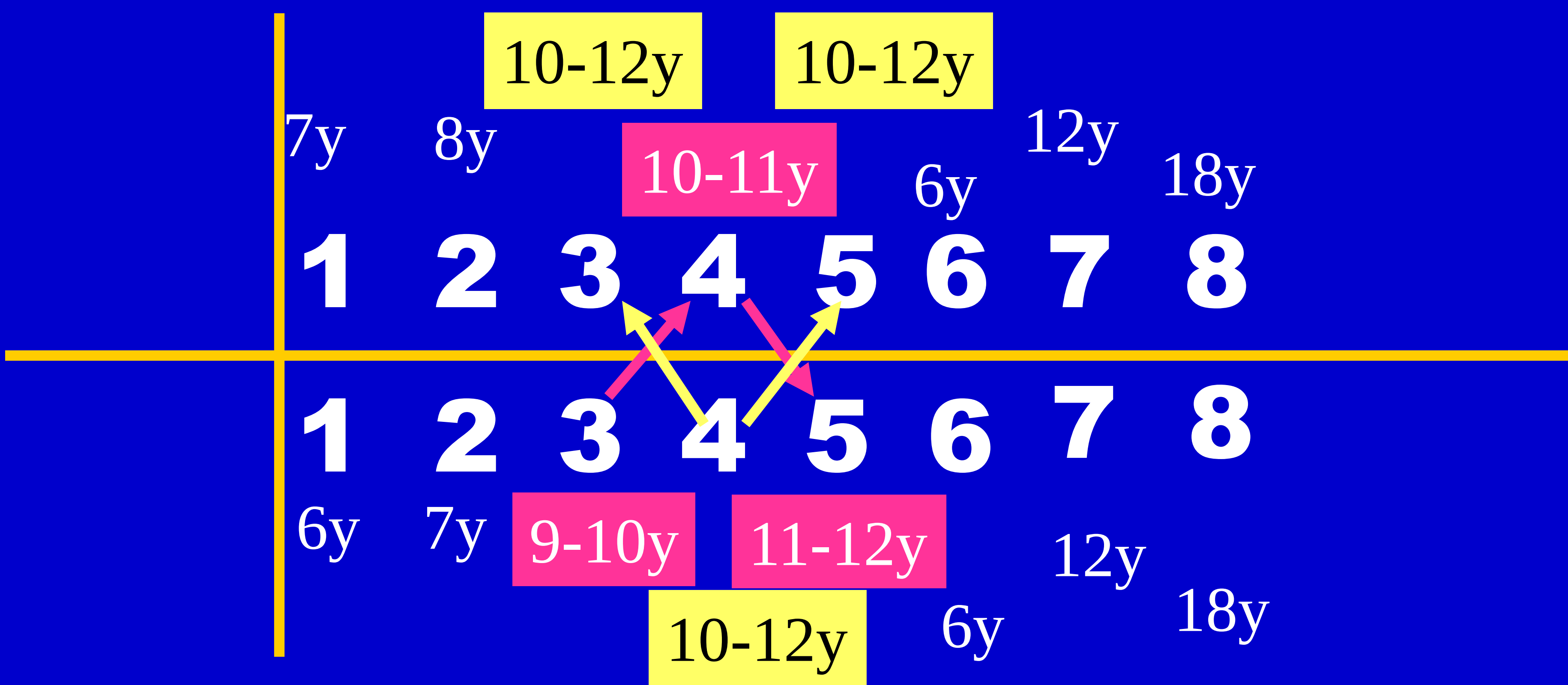
DISTAL

OCCLUSAL

No. of roots:

It has 3 roots

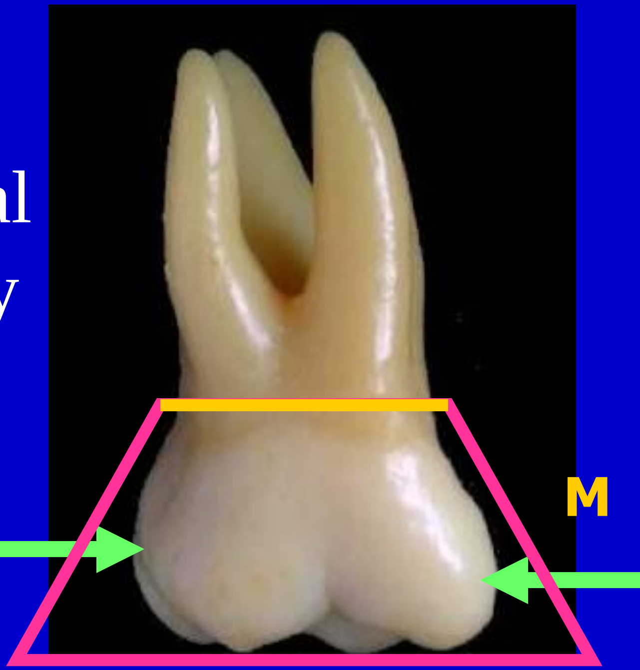
Eruption


10-12y
10-12y
7y
8y
10-11y
6y
12y
18y
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
6y
7y
9-10y
11-12y
12y
10-12y
6y
18y

	Beginning of calc.	Crown completed	Eruption	Root completed
6 6	At birth At birth	-3 years	6-7 y 6-7 y.	+3 years
7 7	2½-3 y. 2½-3 y.	-5 years	12 y. 12 y.	
8 8	7-9 y. 8-10 y.		18 y. 18 y.	

BUCCAL ASPECT

The geometric outline: Trapezoidal with small uneven side present cervically



Mesial outline: Nearly straight till the contact area at the junction of the occlusal and middle 1/3 then convex for the M slope of the MBC.

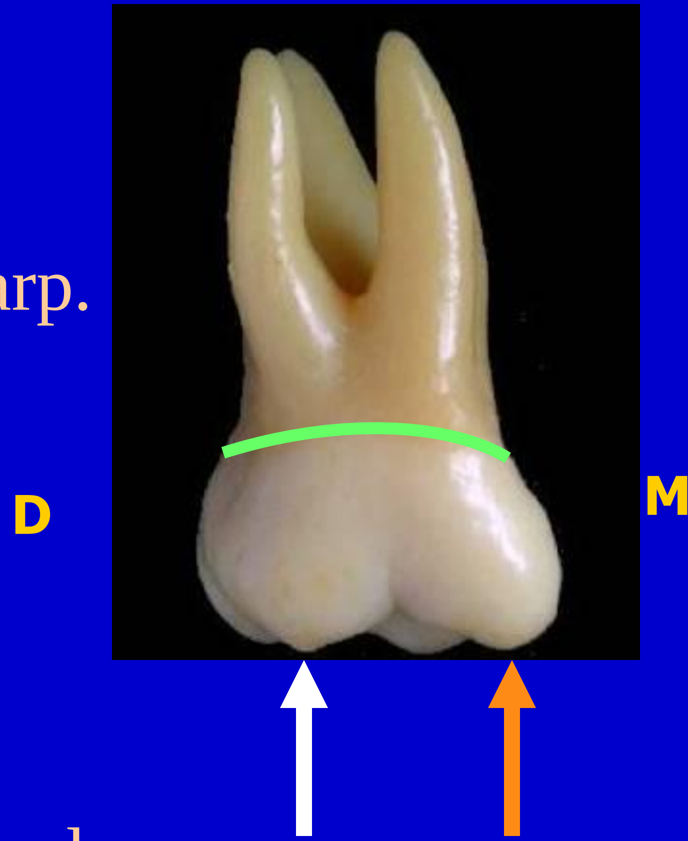
Distal outline: Convex till the contact area at the middle 1/3 then continues convex.

Occlusal outline:

Mesio-buccal cusp is broad and short.

Disto-buccal cusp narrow, long and sharp.

The lingual cusps can be seen due to;
a-Presence of distobuccal convergence.
b-Rhomboidal occlusal outline.



Cervical outline: slightly convex toward the root

Anatomical landmarks:

- **Convex buccal surface.**

***Cervical ridge.**

***Slight depression occlusal to the cervical ridge.**

***Buccal groove may terminate by buccal pit.**

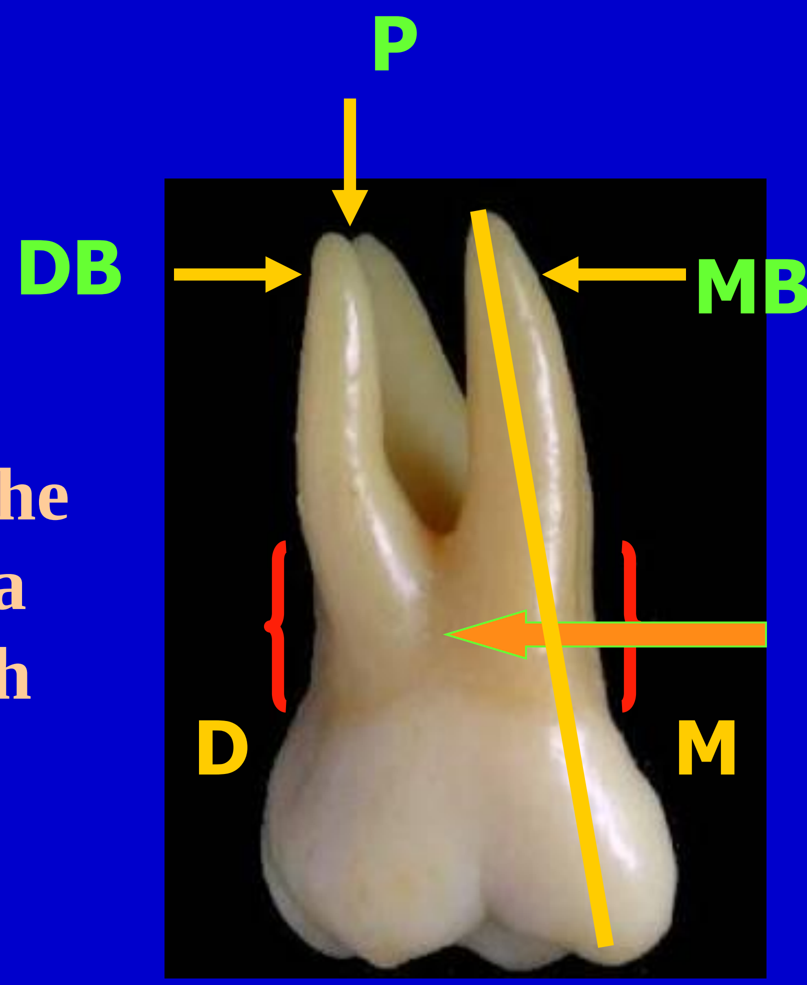
The Roots:

* Root trunk: 4 millimeters

*3 roots are seen .

*MBR curves mesially from the CL till M 1\3,then D&ends in a blunt apex which is in line with MBC.

*DBR is straighter & tends to curve mesially at the apical 1\3



* Deep developmental depression extends on root trunk

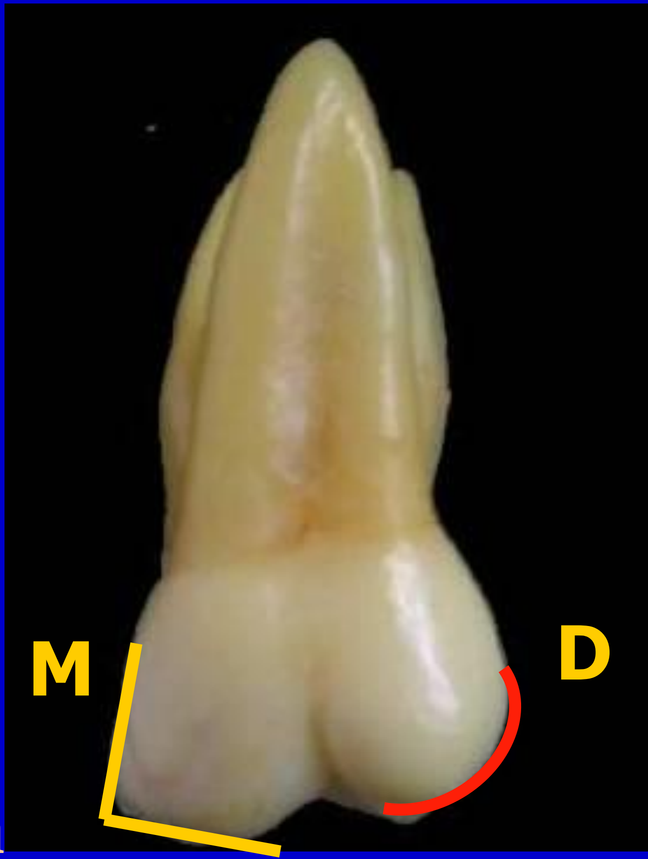
- From this aspect the MB root is *wider and longer* than the DB root. *The palatal root is the longest root from all aspects.

LINGUAL ASPECT

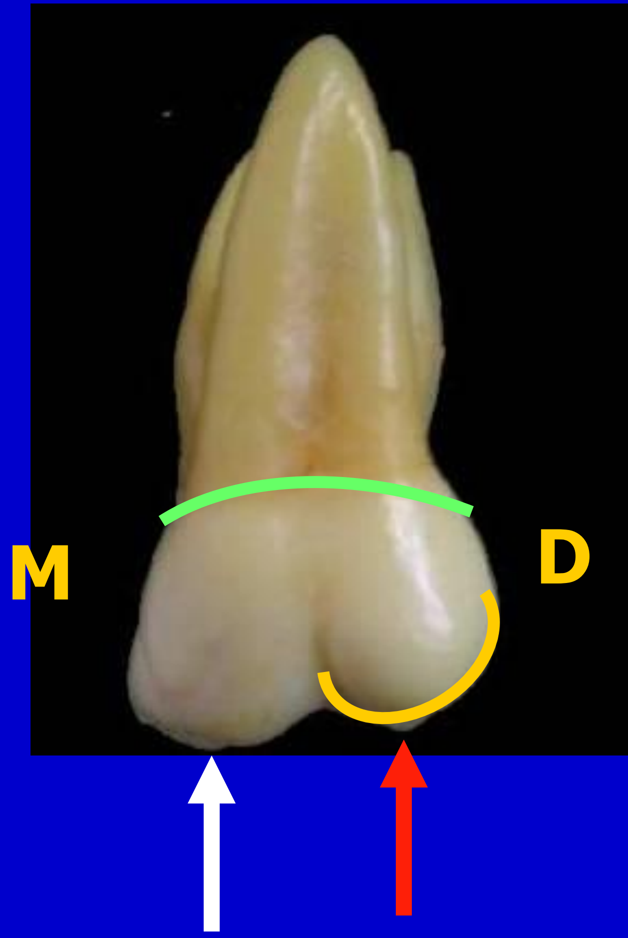
* No lingual convergence

Mesial outline: Nearly straight and form ⊥ with the mesial slope of the ML cusp

Distal outline: Convex and form semicircle with the distal slope of the DL cusp



Occlusal outline: **ML cusp is the largest and longest cusp(3\5 of the crown width). DL cusp is spheroidal**



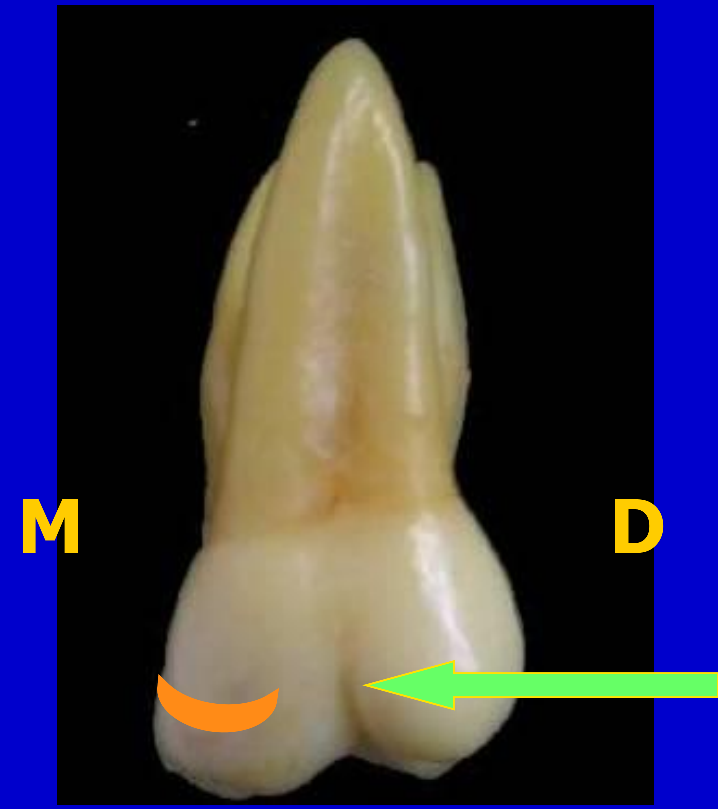
Cervical outline: **Irregular and slightly convex toward the root**

Anatomical landmarks:

*Convex lingual surface.

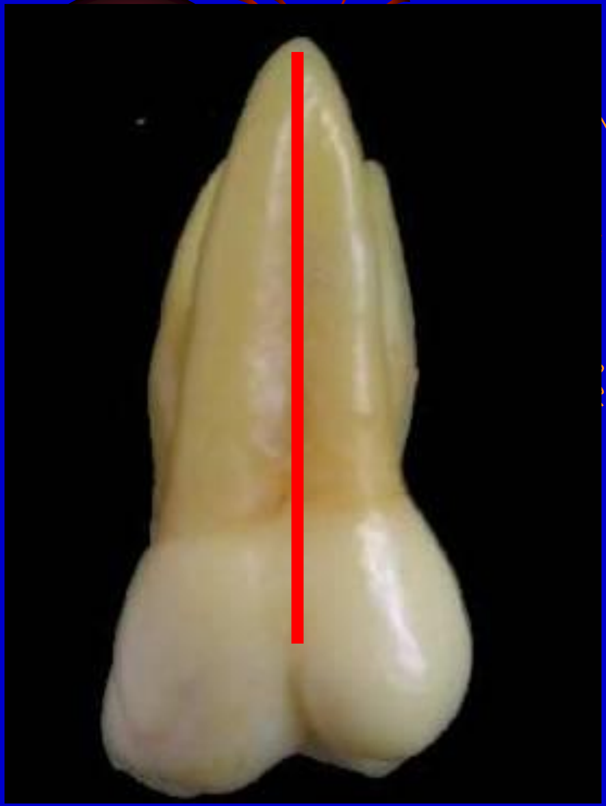
*Tubercle of carabelli on ML cusp (60%).Outlined by a faint DG.

*Lingual developmental groove.



The Roots:

3 roots r seen, the palatal root is the widest root from this aspect.

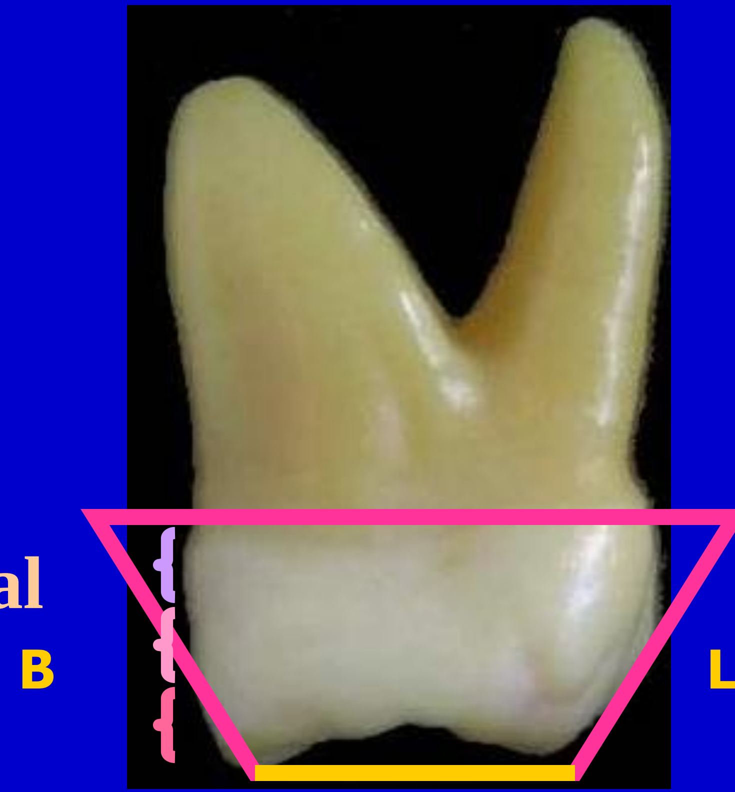


Parts of the buccal roots are seen behind the lingual one.

Its apex is in line with the lingual groove.

MESIAL ASPECT

The geometric outline:
Trapezoidal with small uneven side occlusally



Buccal outline: * Convex at cervical 1/3 denoting cervical ridge

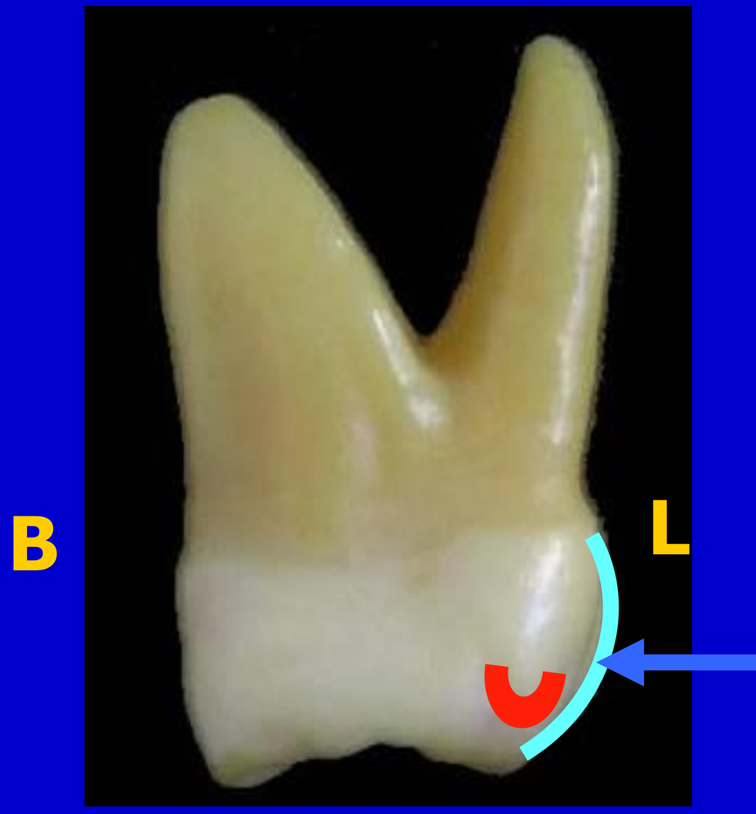
* Concave at the middle 1/3

* Convex at the occlusal 1/3 representing circumscribed MB cusp

Lingual outline:

* Convex with the crest of curvature at the middle 1/3.

* The lingual outline dips inward to illustrate the tubercle.



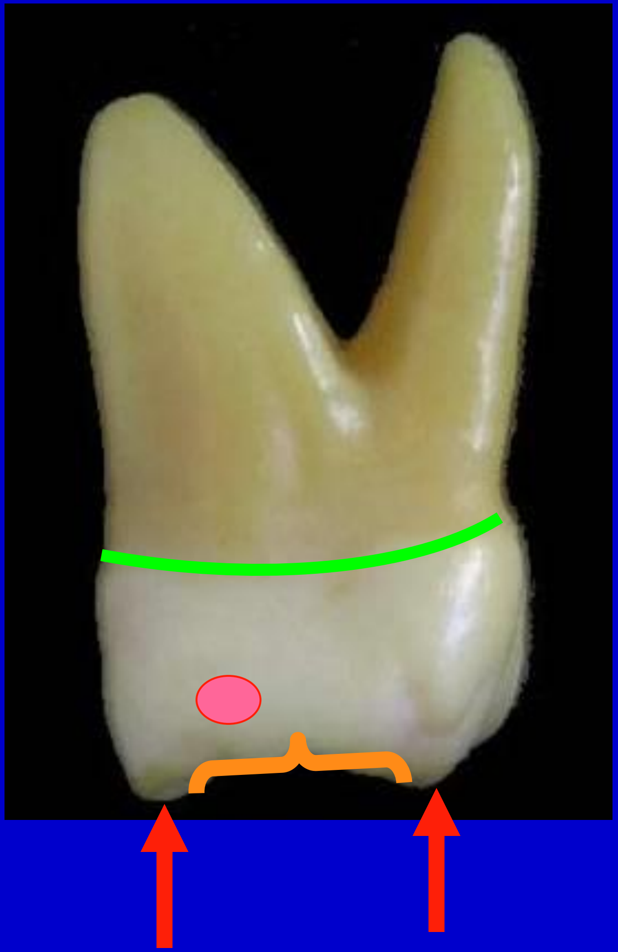
Occlusal outline:

*Represented by ML , MB cusps

*Irregular MMR which curves cervically.

Cervical outline:

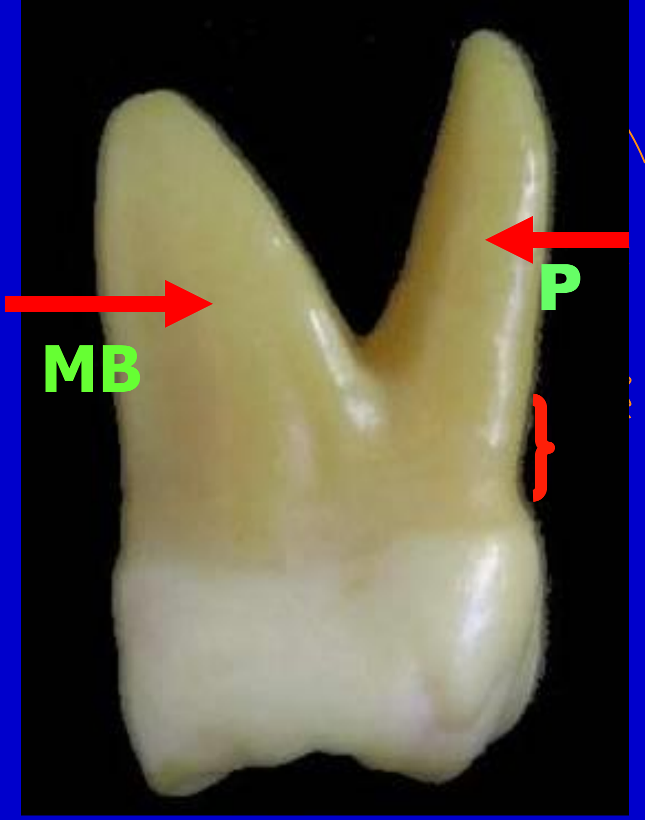
Irregular and convex occlusally



Anatomical landmarks

Mesial contact area at the junction between middle and occlusal 1/3, buccal to the midline

The Roots:



2 roots are seen , MB &palatal.

MB root occupies 2\3 of the BL dimension .
It extends upwards and outwards.

Palatal root is long and narrow; it is banana shaped extending beyond the crown confines.
It has a convex lingual outline extending lingually till the M 1\3 then curves buccally.

Root trunk = 3 mm.

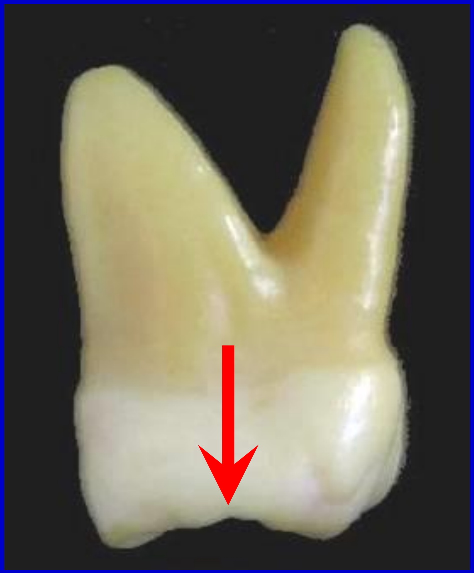
DISTAL ASPECT

*Distal Convergence

*Convex distal surface

*Distal MR curved cervically

MESIAL ASPECT



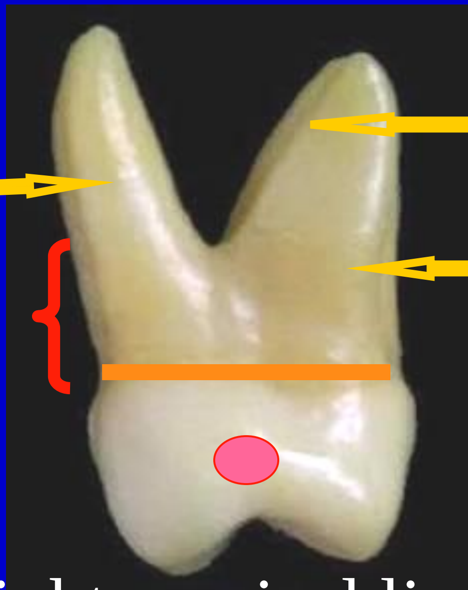
* Wider mesial surface

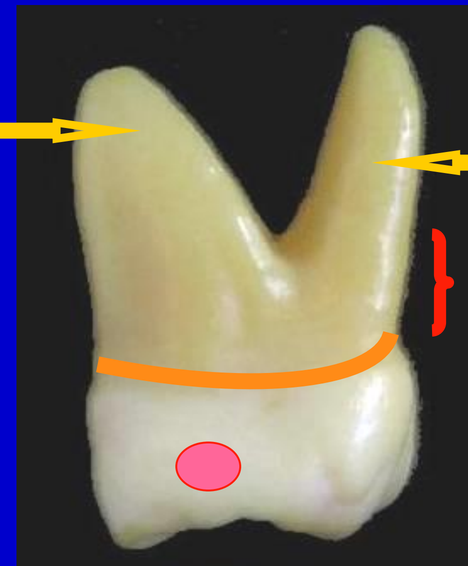
* Flat mesial surface

* Mesial marginal ridge less curved

DISTAL ASPECT

MESIAL ASPECT





*Straight cervical line

*Contact area more cervically, broader

*Root trunk= 5mm

*3 roots can be seen (the MB root is the widest while the P root is the longest)

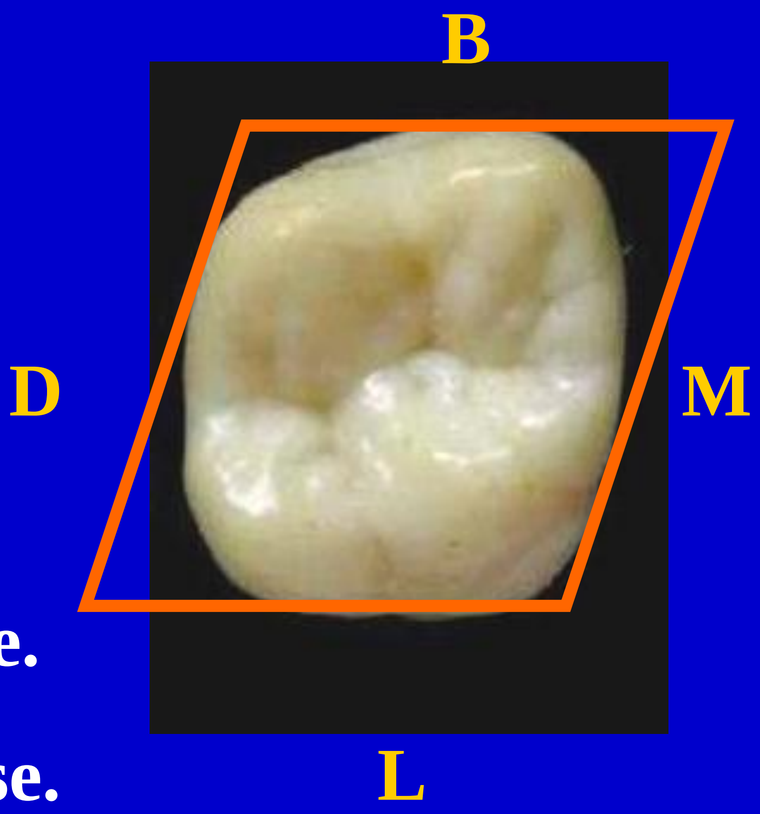
*Cervical line convex occlusally

*Contact area more occlusally

*Root trunk = 3mm

*2 roots can be seen with the same relations

OCCLUSAL ASPECT



The geometric outline:
Rhomboidal

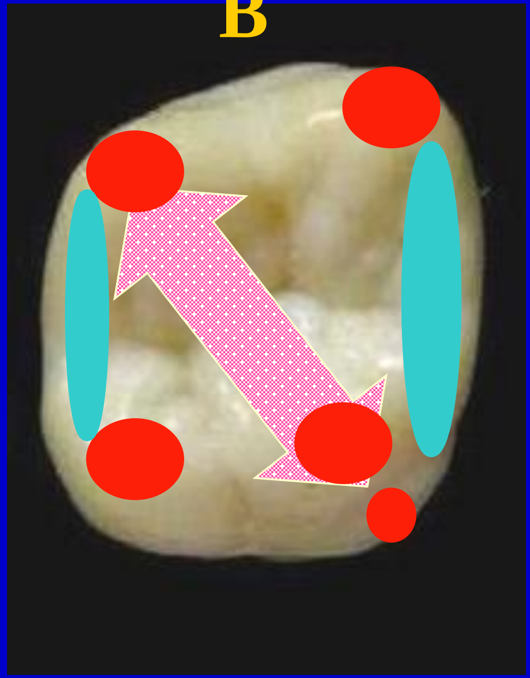
Note: * Disto-buccal convergence.

* ML, BD angles are obtuse.

*MB, DL angles are acute

Anatomical landmarks:

Elevations:



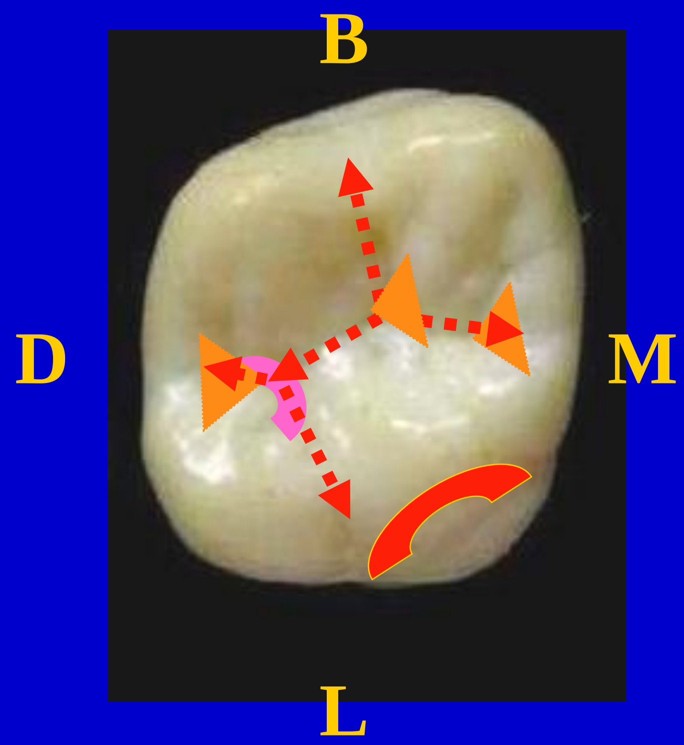
*4 cusps with 4 triangular ridges and tubercle

*Oblique ridge between ML,DB triangular ridge

* MMR and DMR

Anatomical landmarks:

Depressions:



*2 major fossae (central and distal)

*2 Minor fossae (mesial and distal triangular fossae)

*Developmental grooves (Buccal, Central, transverse groove of oblique ridge, distal groove that continues as the lingual groove and fifth cusp groove)

- **Maxillary molar primary cusp triangle theory**
- **Primary cusps= 2 buccal & ML cusps.**

- **Secondary cusps= DLC & cusp of Carabelli.**
- **The arrangement of the 3 major cusps is triangular.**
- **This triangular figure is made up of the 2 buccal cusps ,MMR & the oblique ridge.**

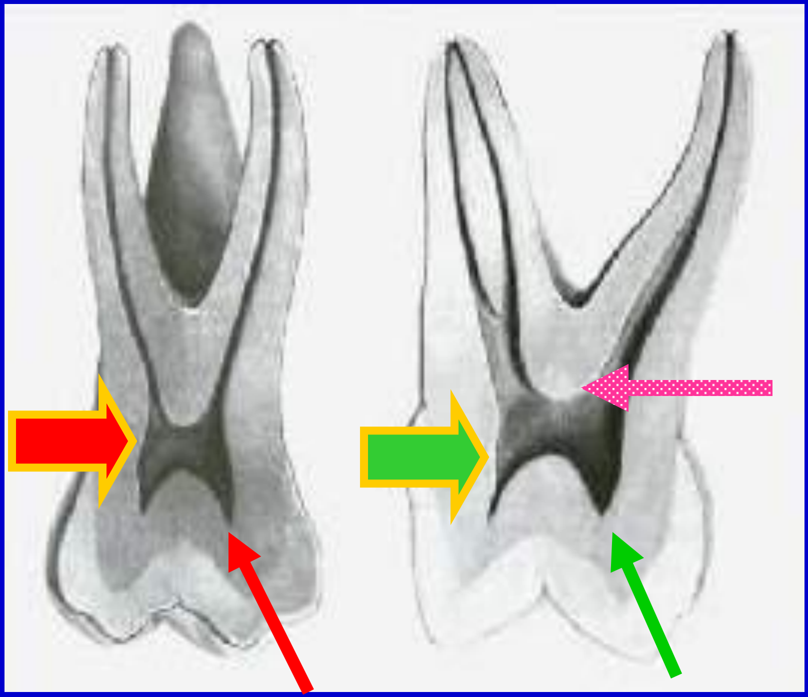
☻ Pulp cavity

►Pulp Chamber:

- The pulp chamber is broader Bucco-lingually than mesio-distally.

-There is pulp horn beneath each cusp

- The floor is apical to the cervical line.

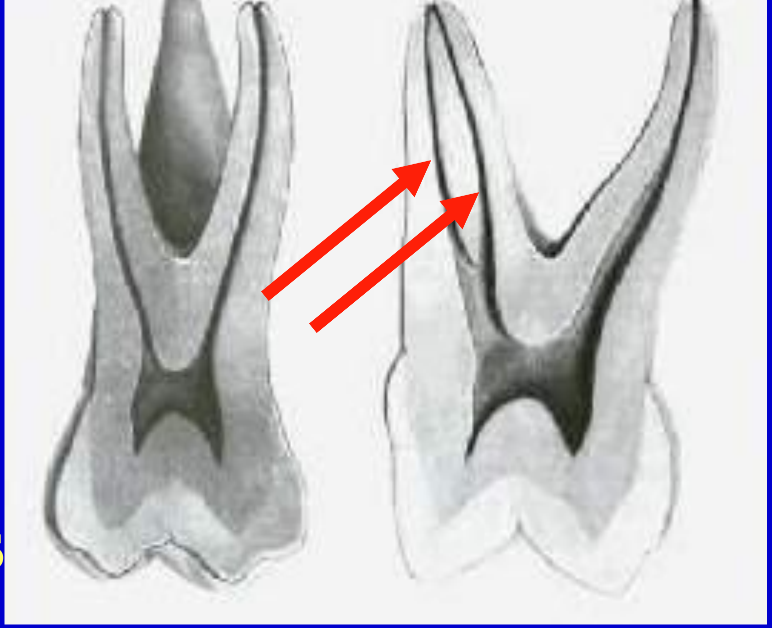


☻ Pulp cavity

►Root Canals:

-3 main root canals.

-MB root may have 2 root canals



Note:

Cervical cross section: Rhomboidal

Mid root section: MB root canal (Oval- Kidney)

DB root canal (Round-Oval)

P root canal (Round- Oval)

Enumerate the names of the following elevations and depression?

1-DMR

2-Oblique ridge

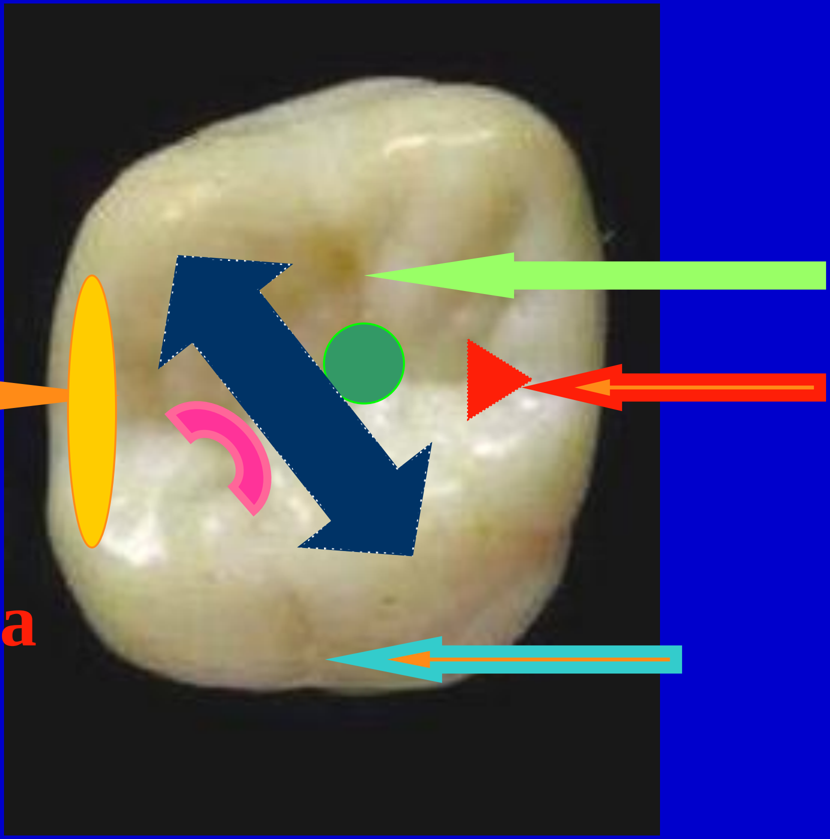
3-Distal linear fossa

4- Central fossa

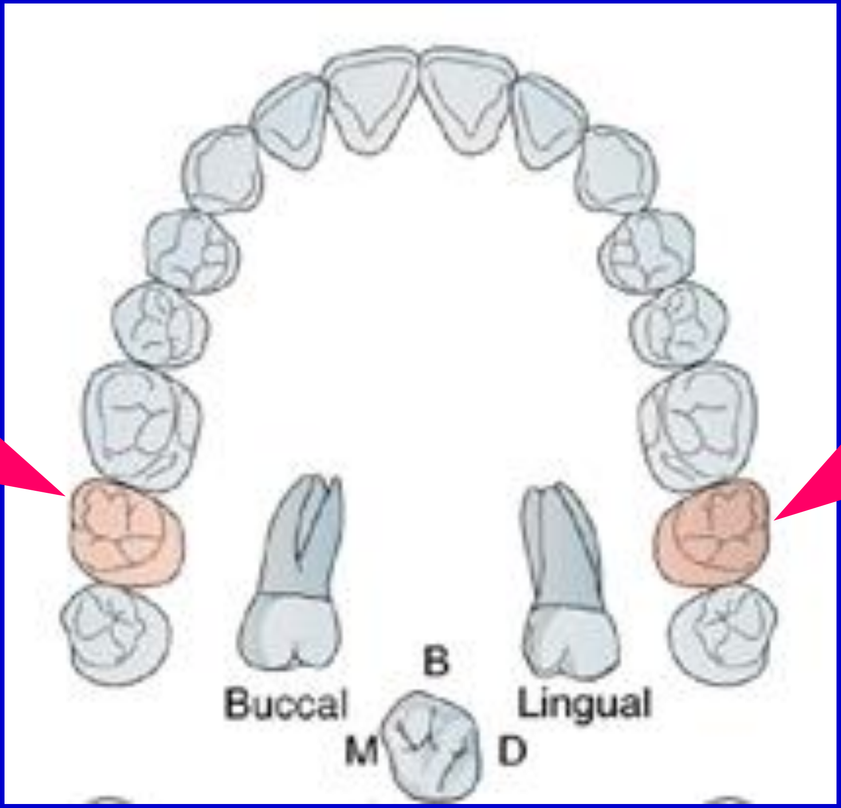
5- Buccal groove

6- Mesial triangular fossa

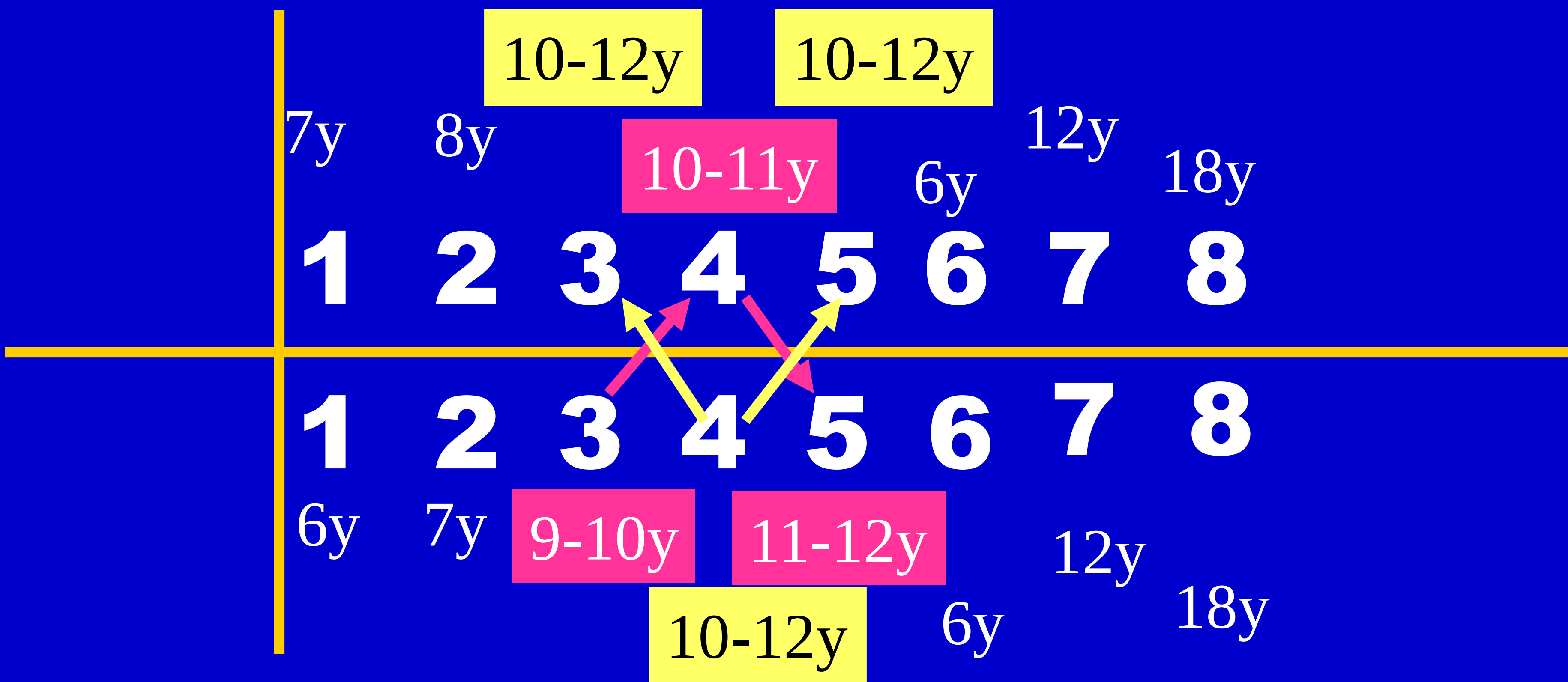
7- Lingual groove



Maxillary permanent 2nd Molar



Eruption

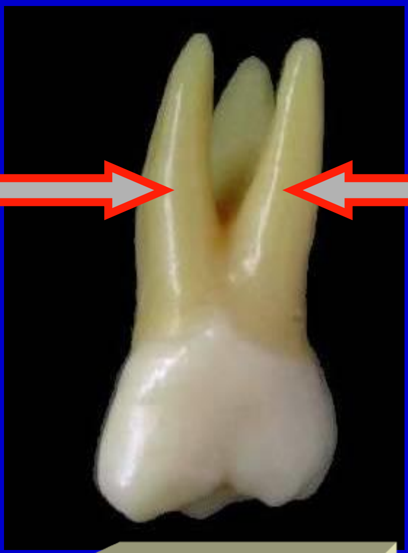


10-12y
10-12y
7y
8y
10-11y
6y
12y
18y
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
6y
7y
9-10y
11-12y
12y
10-12y
6y
18y

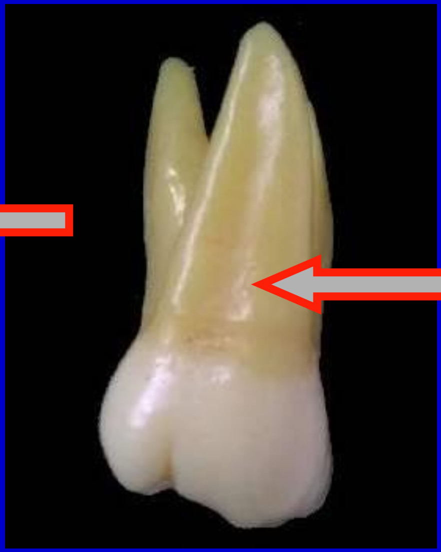
<table>
<tr><th></th><th>Beginning of calc.</th><th>Crown completed</th><th>Eruption</th><th>Root completed</th></tr>
<tr><td>6
6</td><td>At birth
At birth</td><td>-3 years</td><td>6-7 y
6-7 y.</td><td rowspan="3">+3 years</td></tr>
<tr><td>7
7</td><td>2½-3 y.
2½-3 y.</td><td rowspan="2">-5 years</td><td>12 y.
12 y.</td></tr>
<tr><td>8
8</td><td>7-9 y.
8-10 y.</td><td>18 y.
18 y.</td></tr>
</table>

No. of surfaces:

It has five surfaces



BUCCAL



PALATAL

MESIAL

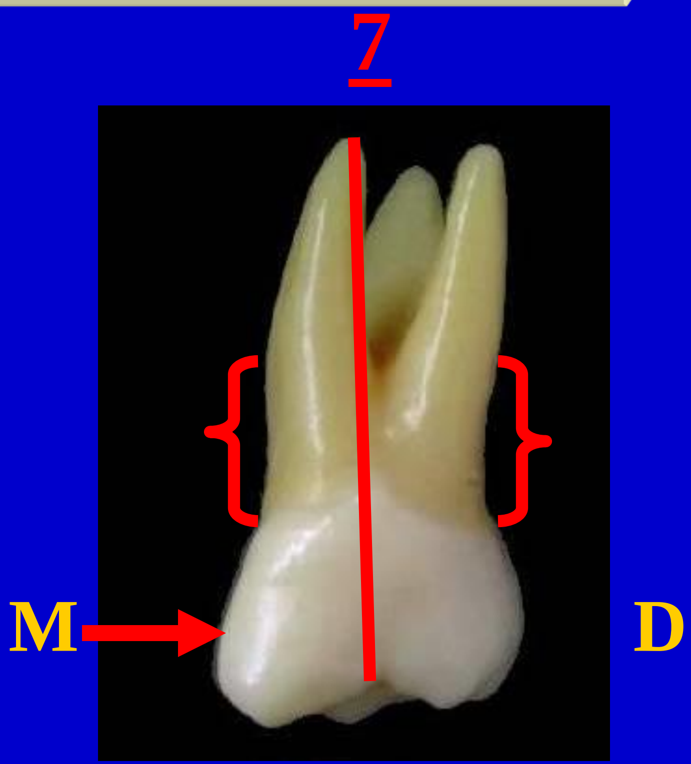
DISTAL

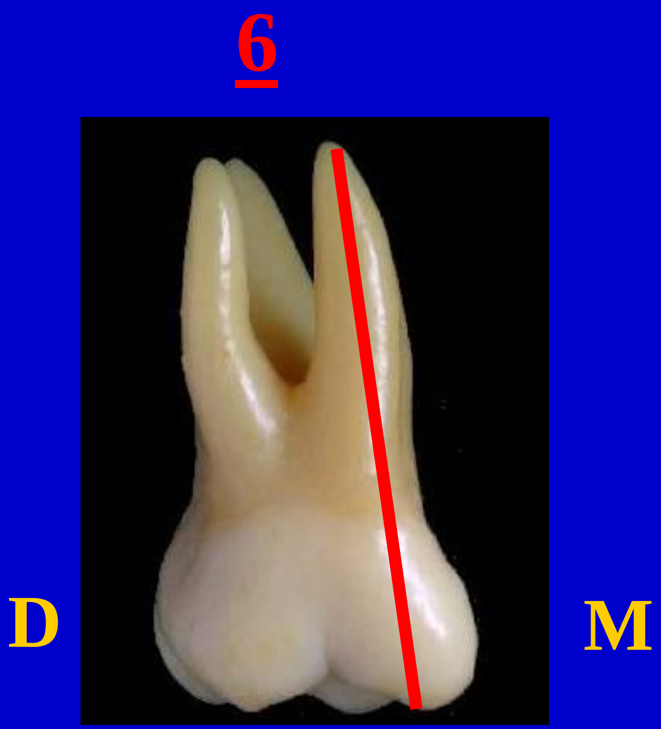
No. of roots:

It has 3 roots

OCCLUSAL

BUCCAL ASPECT





In 7 MB cusp is larger and longer than DB cusp

The root trunk is longer in 7

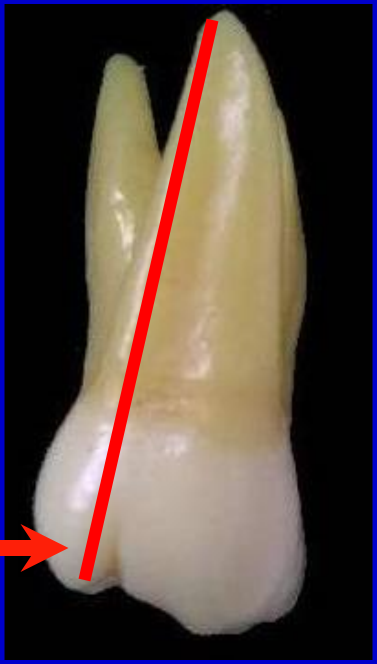
The roots r nearly parallel &more inclined distally

MBR apex is in line with buccal groove

LINGUAL ASPECT

7

D



M

6

D

* DL cusp is smaller in 7 .It may be missed (3 - cusp type)

No tubercle of carabelli in 7•

The apex of the lingual root is in line with the DLC•

MESIAL ASPECT

7

6

* No tubercle of carabelli in 7

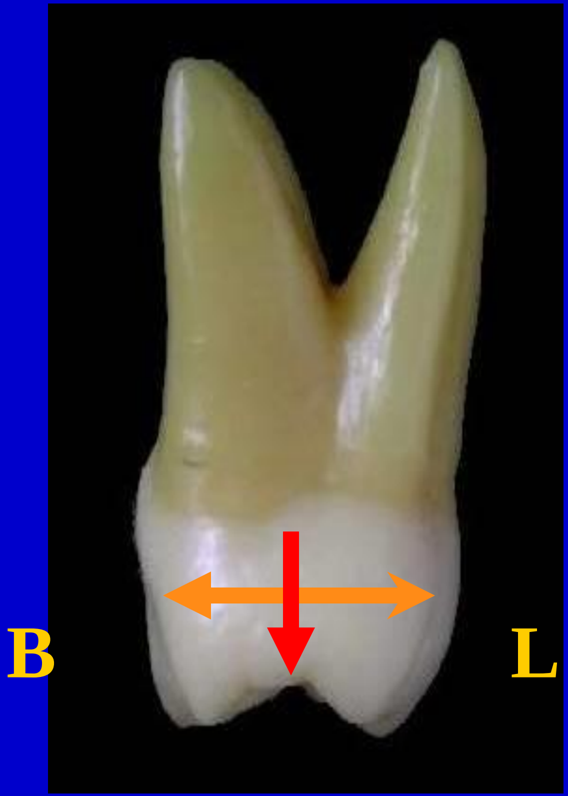
* MB and P roots in 7 are within confines of the BL outlines of the crown

DISTAL ASPECT

* Distal Convergence

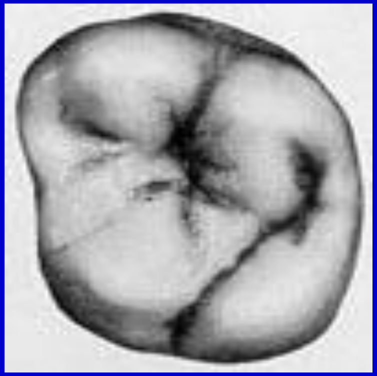
** Convex distal surface

***Distal MR curved cervically



OCCLUSAL ASPECT

B

M  D

L

4 cusp type

outline typical to <u>6</u>

***(obtuse angles r more obtuse &acute r more acute)**

***Disto- lingual convergence**

***No tubercle of carabelli**

B

 M

L

3 cusp type

shape typical to <u>8</u>

***Lingual convergence.**

***LC &DBC r joined by oblique ridge**

***No disto-lingual cusp and no lingual groove.**

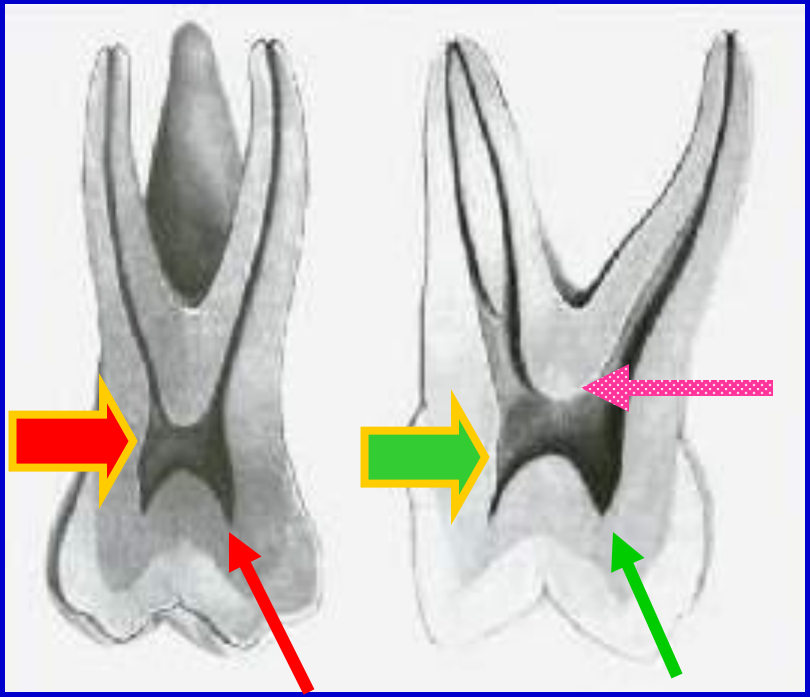
☻Pulp cavity

►Pulp Chamber:

- The pulp chamber is broader Bucco-lingually than mesio-distally.

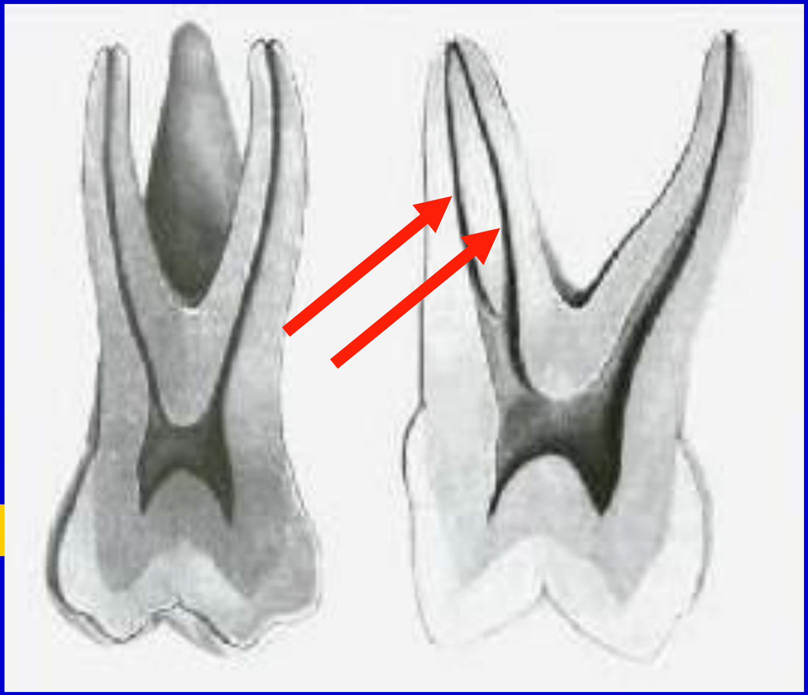
-There is pulp horn beneath each cusp

- The floor is apical to the cervical line.



☻ Pulp cavity

►Root Canals:



-3 main root canals.

-MB root may have 2 root canals

Note:

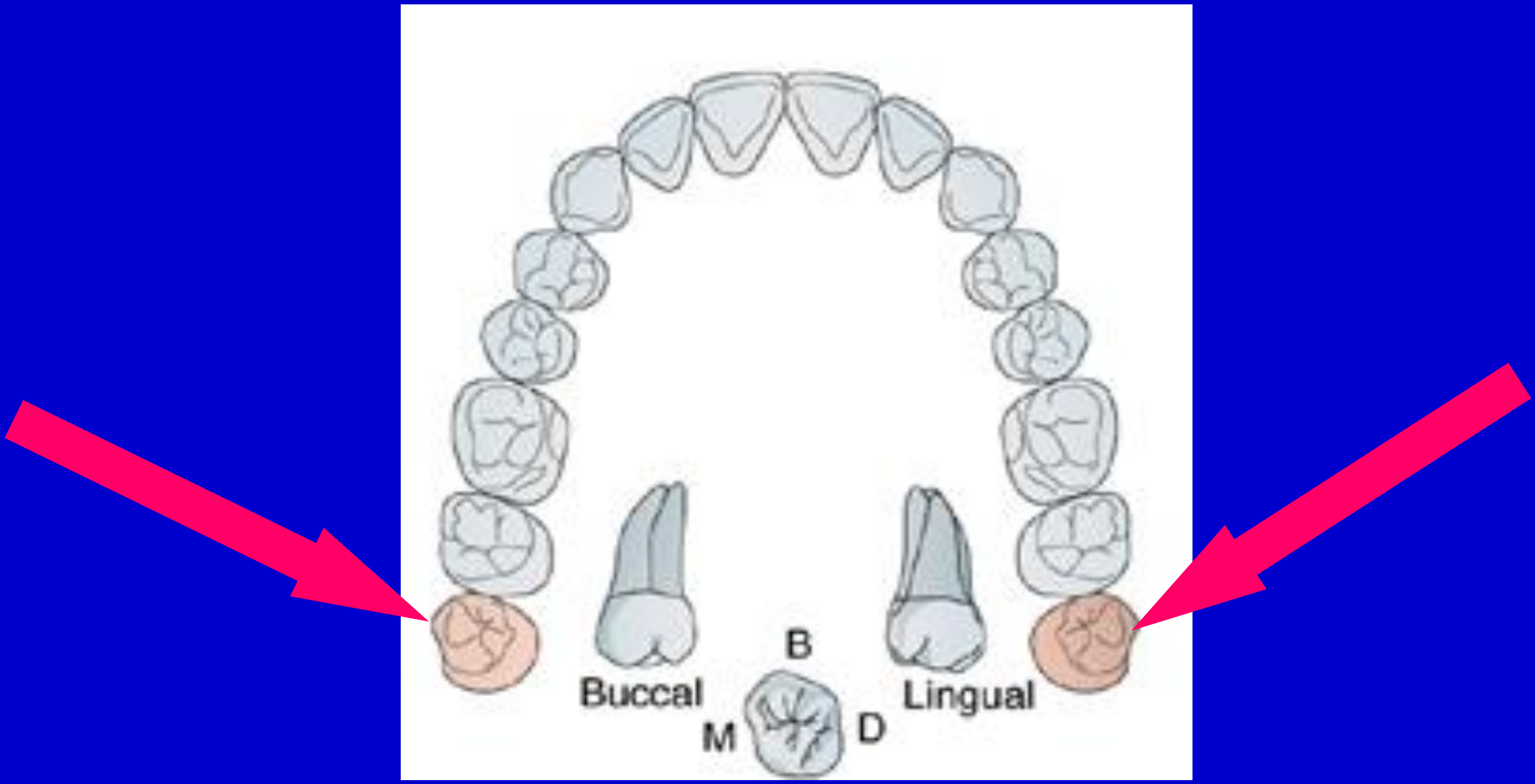
Cervical cross section: Rhomboidal

Mid root section: MB root canal (Oval- Kidney)

DB root canal (Round-Oval)

P root canal (Round- Oval)

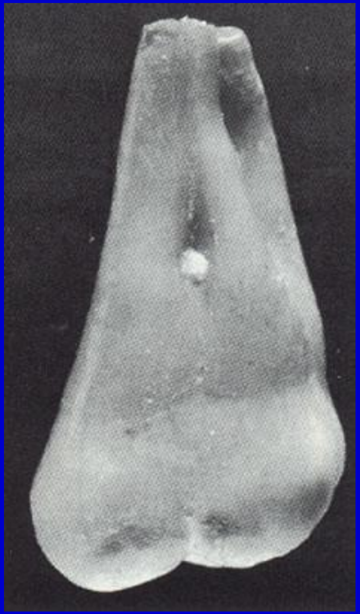
Maxillary permanent 3rd Molar



No. of surfaces:

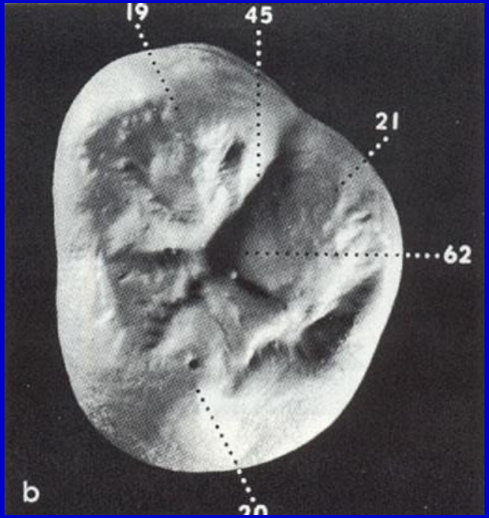
It has five surfaces

BUCCAL

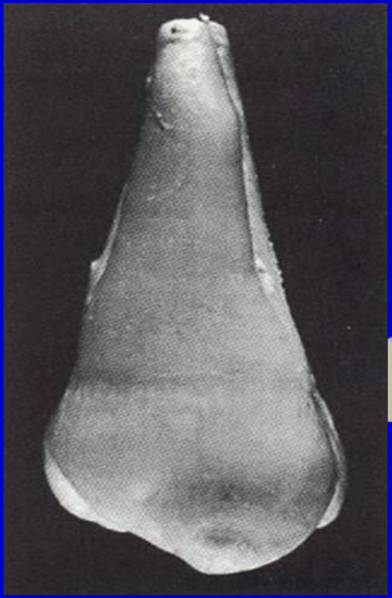


MESIAL

OCCLUSAL



PALATAL

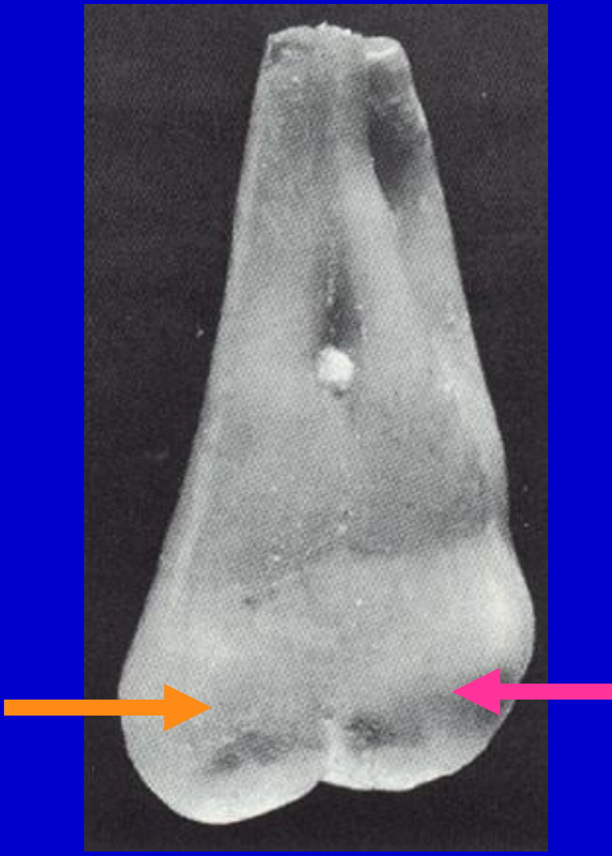


DISTAL

BUCCAL ASPECT

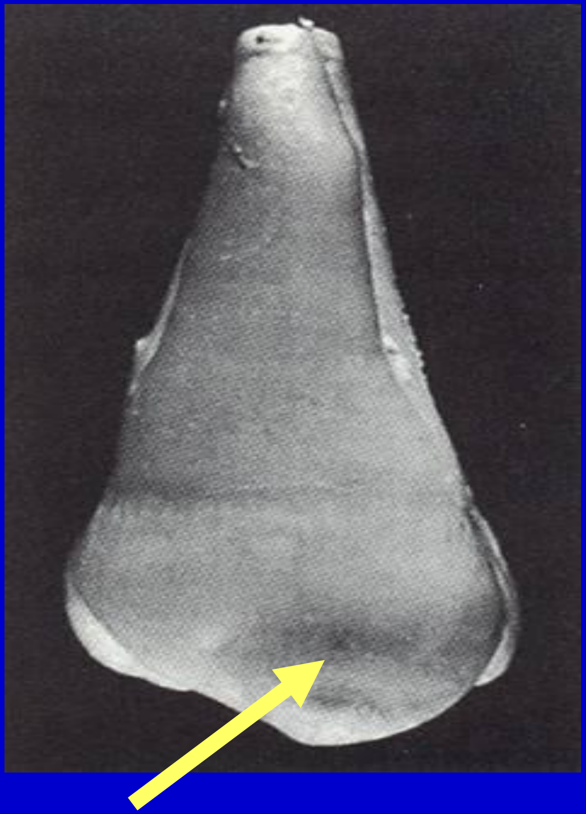
MB cusp is wider and longer than DB cusp

The roots are often fused making one large short root



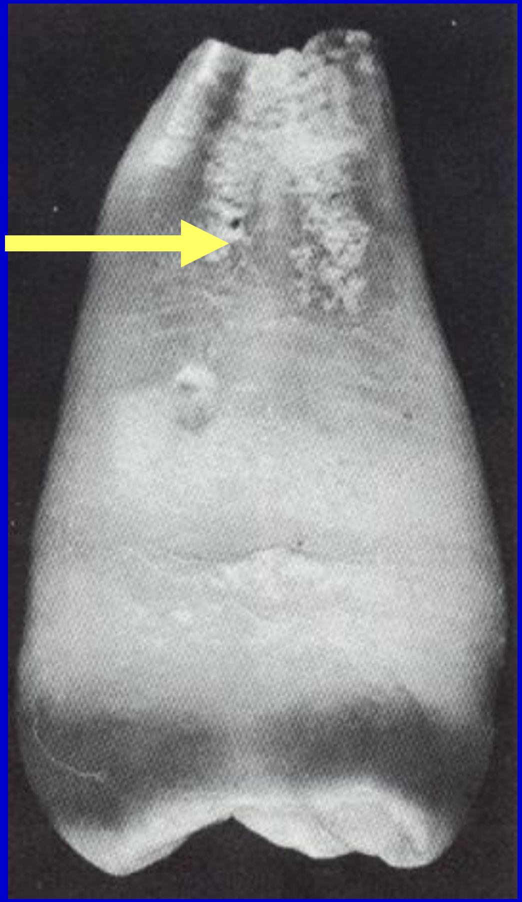
LINGUAL ASPECT

Usually one large lingual cusp
without lingual groove



MESIAL ASPECT

Characterized by fusion of the roots and the bifurcation in the apical 1/3

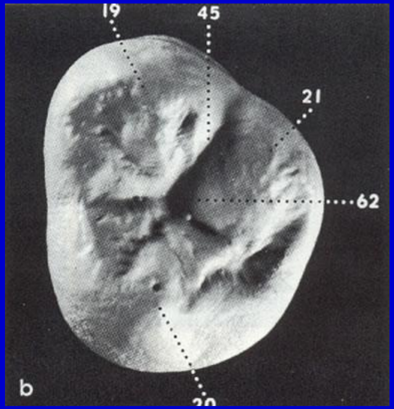


DISTAL ASPECT

*Convex surface with no contact area

*The distal surface short and narrow. So most of the occlusal surface is seen.

OCCLUSAL ASPECT



3 cusp type

♡ **shape is typical to 7**

Lingual convergence,

No lingual groove and no oblique ridge.

4 cusp type

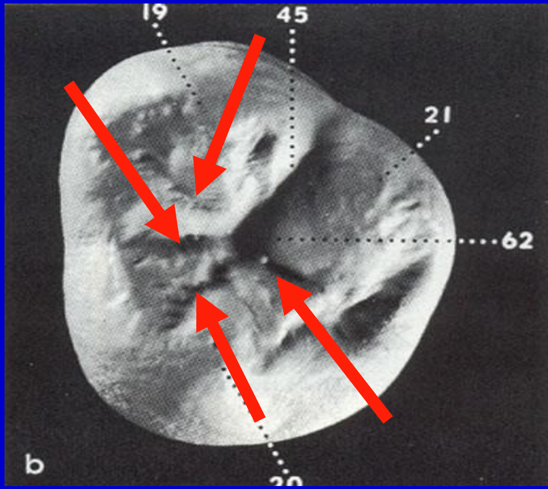
outline typical to 6 with more obtuse angles.

There are disto- lingual cusp, Lingual groove and oblique ridge.

No tubercle of carabelli.

numerous supplemental grooves

OCCLUSAL ASPECT



Notes:

1- The occlusal surface of 8 has numerous supplemental grooves

2- The pulp cavity showed great variation in the size and shape

3- The pulp chamber of 8 is larger than other molars because it is younger than other molars by 9-11 years

Variations

- -So large and malformed.
- -Underdeveloped and conical with small tapering root.
- -Accessory cusps and roots.


GOOD LUCK